

VAJRALA GEETHIKA

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Summary

Highly analytical and detail-oriented engineer enthusiast with a strong academic background and a solid foundation in Relational Database Management Systems (RDBMS) concepts, SQL, and Python programming. Experienced in translating complex business problems into clear technical designs, implementing robust data structures, and developing automated technology solutions. Collaborator who applies Agile methodologies and structured unit testing to ensure high-quality software delivery and seamless integration. Eager to contribute technical problem-solving skills to high-impact client engagements.

Technical Skills

Databases & RDBMS:	SQL, MySQL, Data Extraction, Data Validation, Database Design
Programming & Analytics:	Python, Data Structures, Algorithms, Exploratory Data Analysis (EDA), Statistical Analysis
Reporting & Visualization:	Power BI, Advanced Excel, Data Dashboards
Methodologies & Tools:	Agile, Systems Integration Testing (SIT), Unit Testing, Git/GitHub, VS Code, Jupyter Notebook

PROJECTS

Customer Purchase Behavior Analytics

SQL, Python, Power BI

- Developed an automated reporting dashboard using Python and SQL to extract and process transaction datasets, saving 10 hours of manual effort weekly.
- Created comprehensive performance scorecards in Power BI to track key performance indicators (KPIs) and visualize customer trends.
- Improved database validation procedures, reducing data processing errors by 30% through targeted Python scripting.
- Participated in System Integrated Testing (SIT) and unit testing phases to ensure high reliability and software quality.
- Supported cross-functional client reviews by presenting actionable business insights derived from structured data analysis.

Loan Default Risk Prediction

Python, SQL, Scikit-learn, Pandas

- Developed and validated financial datasets exceeding 50,000 records using Pandas and Python, ensuring data integrity for Management Information Systems (MIS) reporting.
- Implemented SQL queries and Python-based algorithms to analyze credit risk factors, translating complex business requirements into automated data solutions.
- Reduced manual data processing time by 20% by designing and executing optimized SQL scripts for automated data extraction.
- Developed detailed documentation for data structures and reporting models to support internal team knowledge transfer.

AI-Powered Computer Vision Attendance System

Python, OpenCV, SQL

- Developed an automated attendance tracking application using Python and OpenCV, achieving a 95% logging accuracy rate over legacy systems.
- Designed normalized database schemas and implemented RDBMS best practices using SQL to manage data structures for over 1,000 student records.
- Implemented automated unit testing and integrated validation checks, successfully decreasing data processing bottlenecks and exception rates by 15%.
- Executed embedded Quality Assurance (QA) procedures and System Integrated Testing (SIT) to ensure software reliability.

EDUCATION

SASTRA Deemed University | B. Tech - Electronics and Communication Engineering | Oct 2021 - Sep 2025
Thanjavur, Tamil Nadu

CERTIFICATIONS

Microsoft Certified : Python for Data Science

Scaler Certified:SQL & Database Design with MySQL